Lecture 2 (9/16)

Independence, Marginalization, and Conditioning

Question

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$C = 1$ (diagnosed)	0.0021	0.0019
$C = 0$ (not diagnosed)	0.4899	0.5061

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What do we care about?

- Hospital management: What is the chance a random person (M or F) is diagnosed? (marginalization)
- Dr and patient: What is the chance my patient (known to be M or F) will be diagnosed? (conditional probability)

Marginalization

If we don't care about M/F distinction:

$$P(C = 1) = P(C = 1, G = M) + P(C = 1, G = F)$$
$$= 0.0021 + 0.0019 = 0.004$$

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We used that probability of disjoint events as: **(Countable) additivity**: for any countable collection (meaning we can list them as $E_1, E_2, \ldots$) of pairwise disjoint sets $E_1, E_2, \ldots$,

$$P\Big(\bigcup_i E_i\Big) = \sum_i P(E_i).$$

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Marginalization: Sum out the variable you don't care about:

$$P_X(x) = \sum_y P(x, y).$$

Independence

Random variables X, Y are **independent** if

$$P_{X,Y}(x, y) = P_X(x)\, P_Y(y) \qquad \text{for all } x, y.$$

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$$P(X = x, Y = y) = \begin{cases} 1/2 & (0,0) \\ 1/8 & (0,1) \\ 1/8 & (1,0) \\ 1/4 & (1,1) \end{cases}$$

Are they independent?

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$P(X, Y)$	$Y = 0$	$Y = 1$
$X = 0$	$1/8$	$3/8$
$X = 1$	$1/8$	$3/8$

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How many checks for a 2×2 table?

Let X and Y have (marginal) distributions

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Suppose

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Does this mean X and Y are independent?
The other three cells are not free: rows and columns must sum to the marginals, so

$$\begin{aligned} P(X = 1, Y = 0) + P(X = 1, Y = 1) &= P(X{=}1) = p \\ \implies P(X = 1, Y = 0) &= p - P(X = 1, Y = 1) \\ &= p(1 - q) = P(X = 1)P(Y = 0) \end{aligned}$$

Same for the other entries. In general: for variables X and Y with n and m outcomes respectively, we need to check $(m-1)(n-1)$ entries.

Question

$X \in \{0, 1\}$ and $Y \in \{1, 2, 3, 4\}$ have joint distribution

$P(X, Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$Y = 4$
$X = 0$	$1/16$	$1/16$	$1/16$	$1/16$
$X = 1$	$3/16$	$3/16$	$3/16$	$3/16$

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$X = 1$	0	$1/8$	$1/8$	$1/4$

Are X and Y independent?

Cancer diagnosis data: compare the joint probabilities to the products of the marginals:

(C, G)	Joint $P(C, G)$	Product $P(C)P(G)$
$(1, M)$	0.0021	0.001968
$(1, F)$	0.0019	0.002032
$(0, M)$	0.4899	0.490032
$(0, F)$	0.5061	0.505968

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- If we can't do this, we need to quantify statistical uncertainty (Unit 3)
- Are men more or less likely to be diagnosed?

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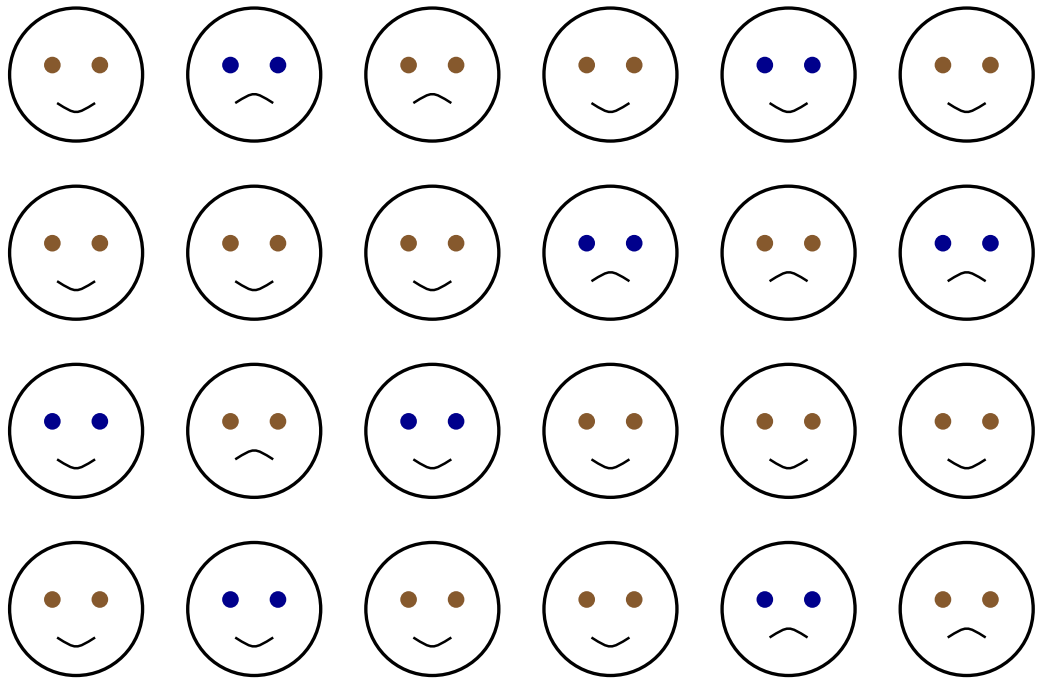
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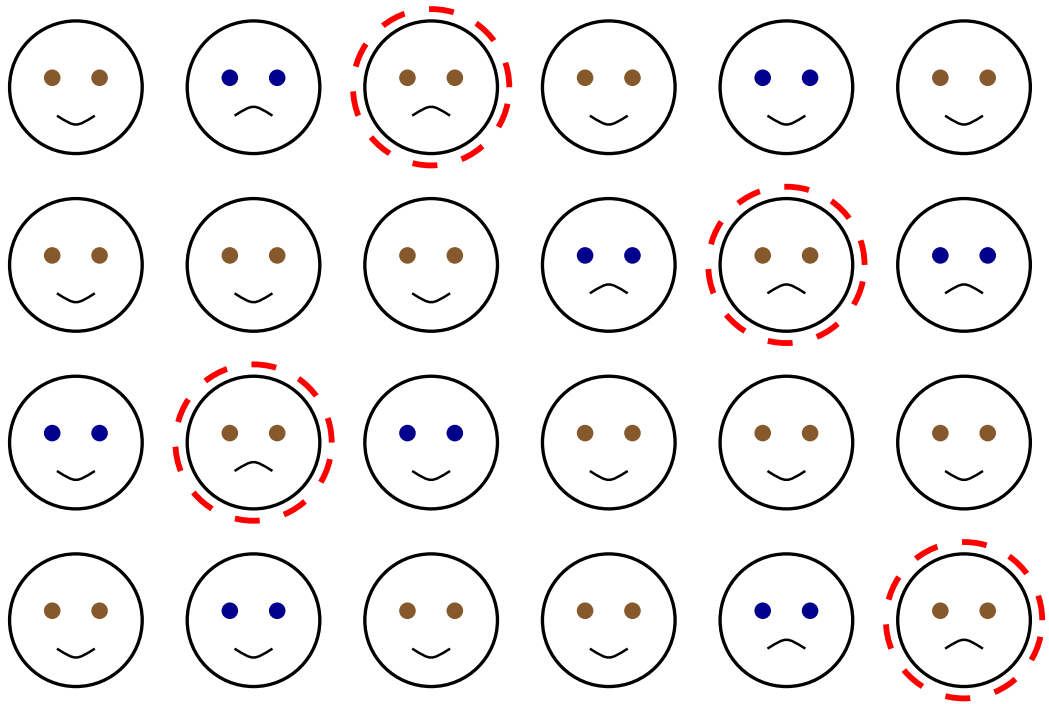
$P(x \mid y)$ is the fraction of times x is observed among outcomes for which y is observed.

- Marginalize over X means: I no longer see X
- Conditioning on X: I fix X to a certain value

Joint, marginal, and conditional probability: a population



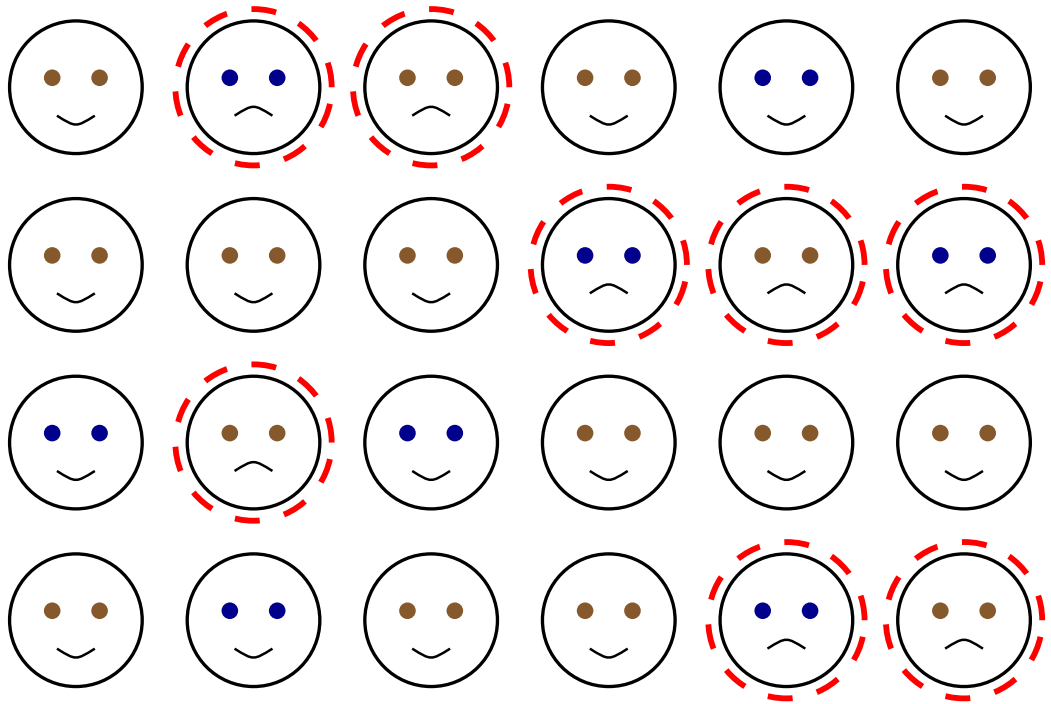
Joint, marginal, and conditional probability: a population



A **joint probability**: brown eyes *and* sad.

$$P(\text{brown}, \text{sad}) = \frac{4}{24} = \frac{1}{6}.$$

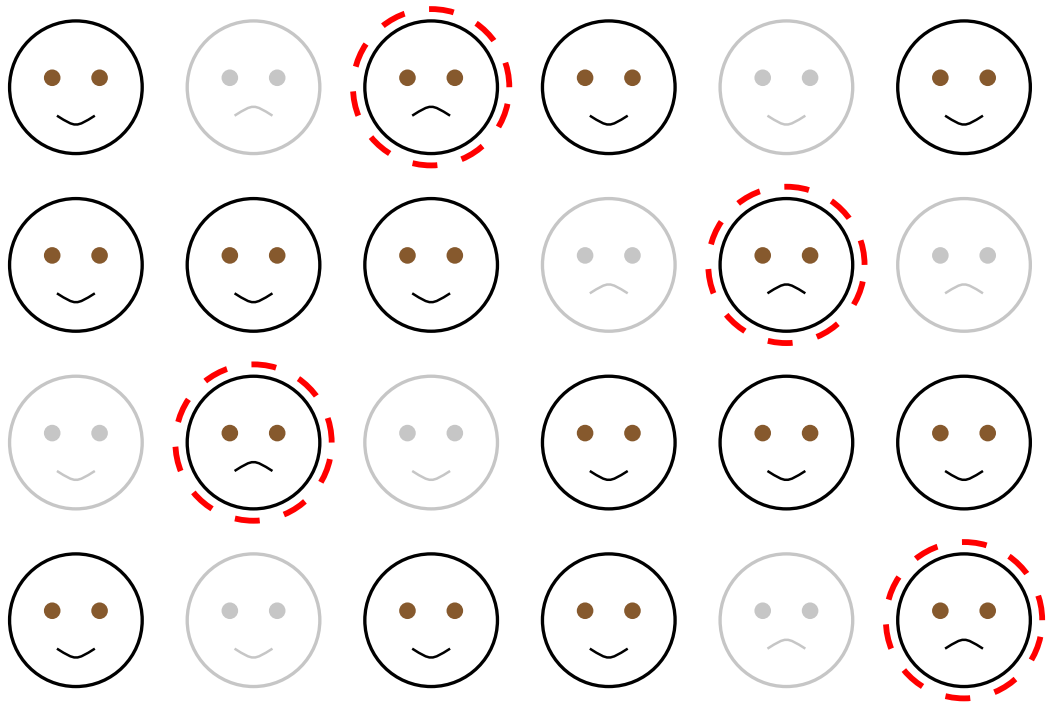
Joint, marginal, and conditional probability: a population



Marginalizing over eye color.

$$P(\text{sad}) = \frac{8}{24} = \frac{1}{3}.$$

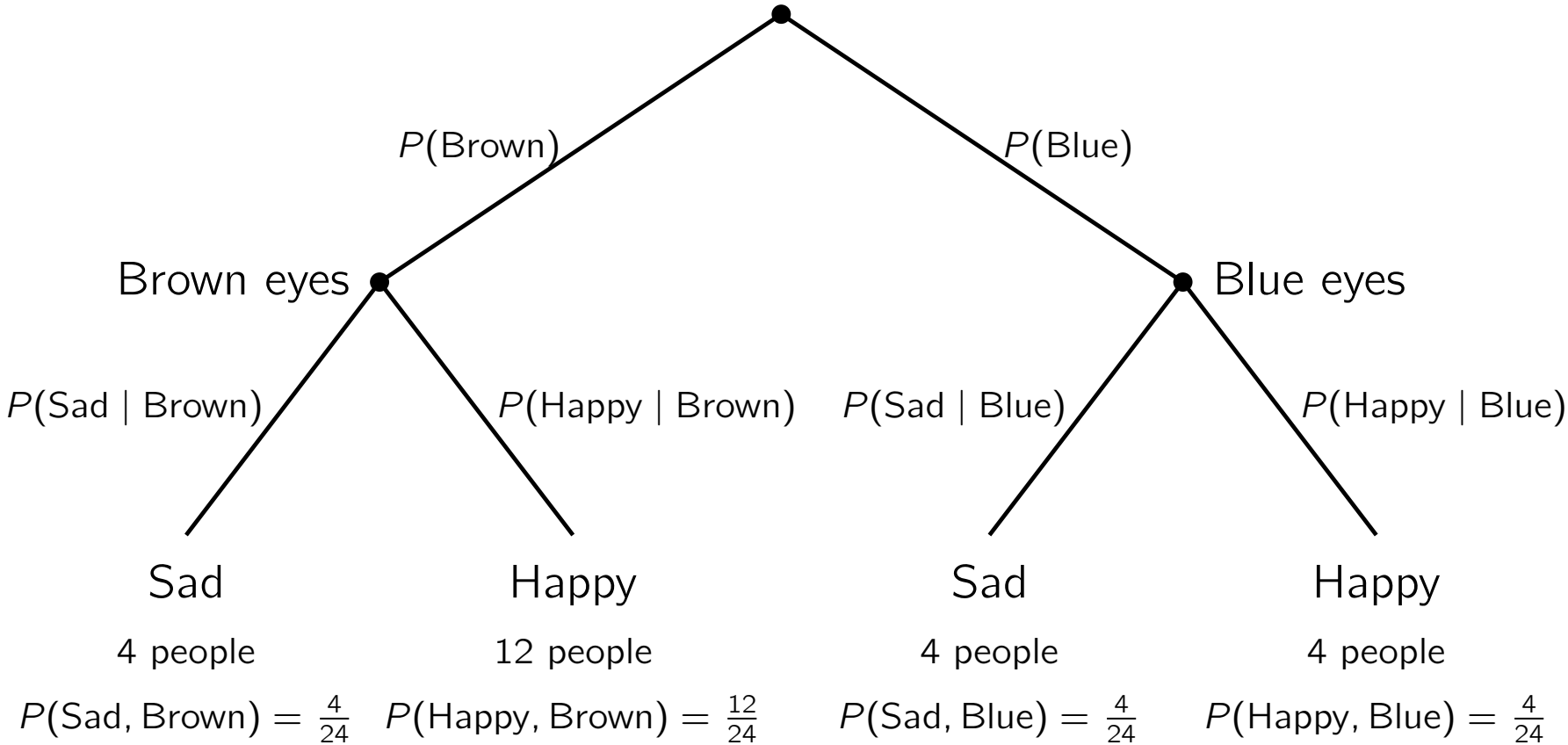
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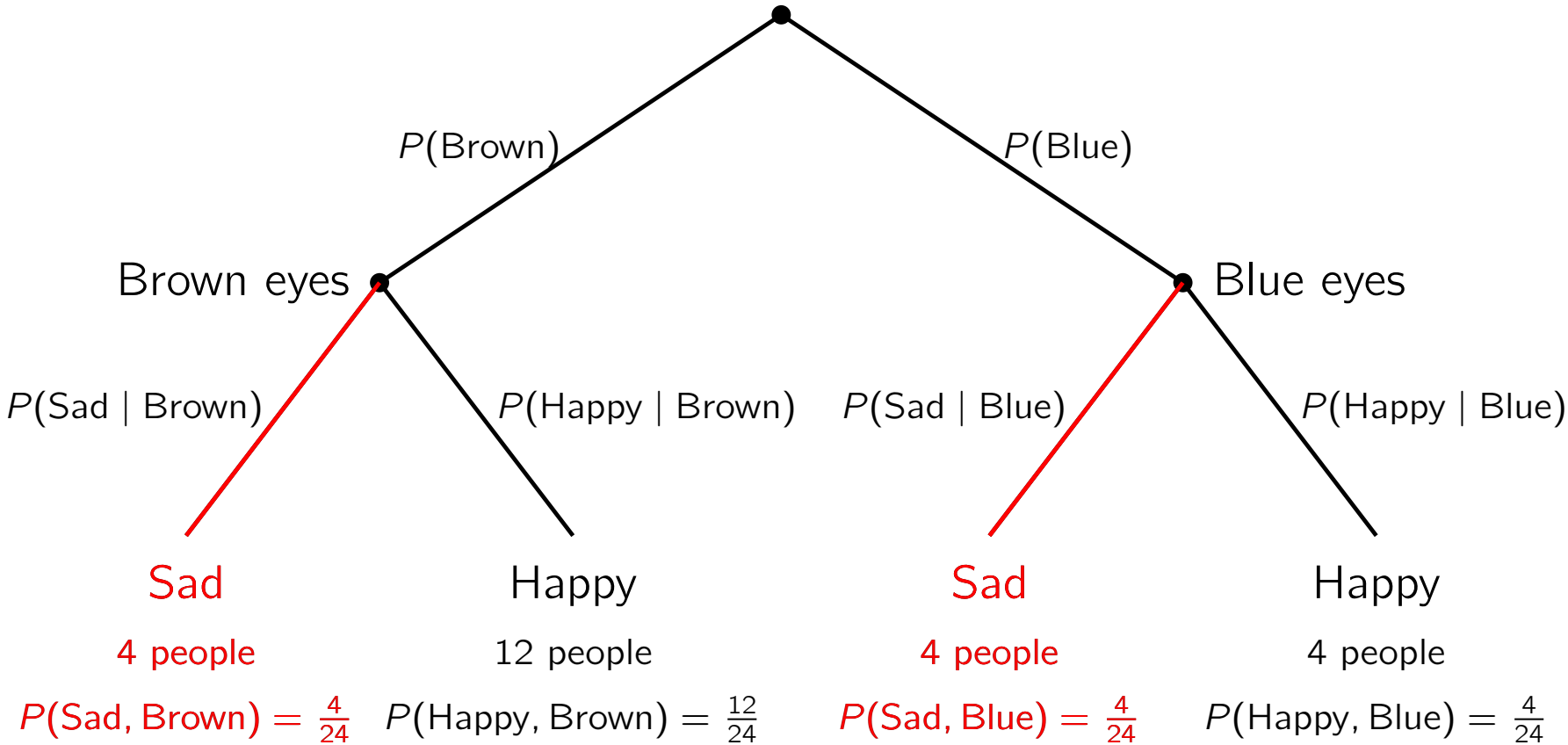
Conditioning on brown eyes.

$$P(\text{sad} \mid \text{brown}) = \frac{4}{16} = \frac{1}{4}.$$

Marginalizing vs. conditioning: a tree



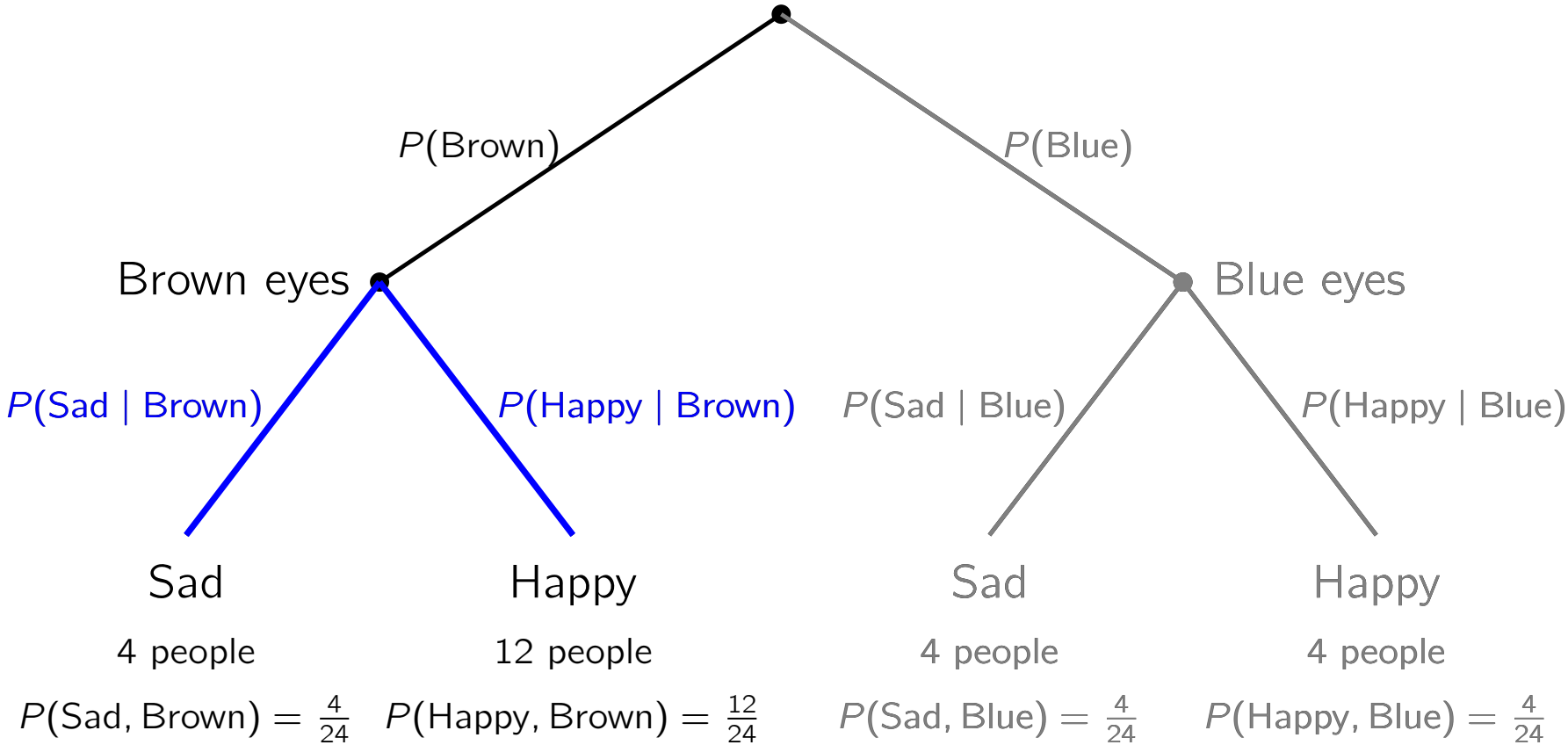
Marginalizing vs. conditioning: a tree



Marginalizing over eye color: lump the two highlighted branches together.

$$P(\text{sad}) = \underbrace{\frac{4}{24}}_{\text{sad, brown}} + \underbrace{\frac{4}{24}}_{\text{sad, blue}} = \frac{8}{24} = \frac{1}{3}.$$

Marginalizing vs. conditioning: a tree



Conditioning on brown eyes: forget the blue-eyed branch, read off proportions within what's left.

$$P(\text{sad} \mid \text{brown}) = \frac{4}{16} = \frac{1}{4}.$$

Question

$X, Y \in \{0, 1\}$ have joint distribution

$$P(X = x, Y = y) = \begin{cases} 3/8 & (0,0) \\ 1/8 & (0,1) \\ 1/4 & (1,0) \\ 1/4 & (1,1) \end{cases}$$

Find $P(Y = 1 \mid X = 0)$.

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Find $P(Y = 1 \mid X = 0)$.

Question

Now suppose $X \in \{0, 1\}$ and $Y \in \{1, 2, 3, 4\}$ have joint distribution

$P(X, Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$Y = 4$
$X = 0$	$1/8$	$1/8$	$1/4$	$1/8$
$X = 1$	$1/8$	0	$1/8$	$1/8$

Find $P(X = 1 \mid Y = 3)$.

Example

Cancer diagnosis data again

$C \mid G$	Marginal $P(C)$	Conditional $P(C \mid G)$
$1 \mid M$	0.0040	0.00427
$1 \mid F$	0.0040	0.00374
$0 \mid M$	0.9960	0.9957
$0 \mid F$	0.9960	0.9963

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- Are they independent? Same problem as marginal table.
- But it is much easier to see the dependence:

$$\text{effect of being male} = P(1|M) - P(1|F) = 0.00427 - 0.00374 = 0.00053$$

Taste of regression models

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- Thus it is natural to consider models given to us in conditional form: For a pair of Bernoullis (X, Y), instead of defining a joint distribution, we can write

$$\begin{aligned} X &\sim \text{Bernoulli}(q_X) \\ Y|X &\sim \text{Bernoulli}(q_1 X + q_0(1-X)) \quad q_1, q_0 \in [0,1] \end{aligned}$$

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- $Y|X$ means the distribution of Y conditioned on a value of X
- In general, a regression model takes the form

$$Y|X \sim \text{Name}(\text{Parameters})$$

where now "Parameters" depends on X (if they are not independent).

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Example

From the cancer diagnosis data we saw (coding $G = 1$ for male, $G = 0$ for female)

$$q_1 = P(C = 1 \mid G = 1) = 0.00427, \qquad q_0 = P(C = 1 \mid G = 0) = 0.00374.$$

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- $0.00053 = q_1 - q_0$ is the "effect" of being male on the diagnosis probability.

Question

Consider

$$P(X = x, Y = y) = \begin{cases} 1/2 & (0,0) \\ 1/8 & (0,1) \\ 1/8 & (1,0) \\ 1/4 & (1,1) \end{cases}$$

Write a conditional model for $Y|X$.

Question

$$X \sim \text{Bernoulli}(1/3)$$
$$Y|X \sim \text{Bernoulli}(X/2 + 1/3)$$

Find all the joint probabilities and the marginal of Y.

Condition on an even involving a sequence

Question

We are betting on a series of events, each modeled as independent Bernoulli variable $X_i \sim \text{Bernoulli}(q)$ for $i = 1, 2, 3, \ldots$. Let[a]

$$K = \min\{i : X_i = 0\}$$

- Find $P(K = 4)$
- Find $P(K = 10 | K > 6)$.

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- Find $P(K = 4)$
- Find $P(K = 10|K > 6)$.

$$P(K = 4) = P(X_1 = X_2 = X_3 = 1, X_4 = 0) = q^3(1 - q)$$

$$P(K = 10 \mid K > 6) = \frac{P(K = 10)}{P(K > 6)} = \frac{q^9(1 - q)}{q^6} = q^3(1 - q)$$

[a] We say $K \sim \text{Geometric}(q)$, but you don't need to remember this.

A Markov Chain: conditioning on a history

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$$X_0 \sim \text{Bernoulli}(1/2)$$
$$X_{t+1} \mid (X_t, X_{t-1}, \ldots, X_0) \sim \text{Bernoulli}(q_1 X_t + q_0(1 - X_t))$$

and assume $X_0 = 1$. This says $X_{t+1} \mid X_t$ is independent of $X_{t-1}, \ldots, X_0$, since those don't appear in the Bernoulli random variable.

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Let $p_t = P(X_t = 1)$.

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and assume $X_0 = 1$. This says $X_{t+1} \mid X_t$ is independent of $X_{t-1}, \ldots, X_0$, since those don't appear in the Bernoulli random variable.
Let $p_t = P(X_t = 1)$. Then

$$\begin{aligned} p_t &= P(X_t = 1 \mid X_{t-1} = 1)P(X_{t-1} = 1) + P(X_t = 1 \mid X_{t-1} = 0)P(X_{t-1} = 0) \\ &= q_1 p_{t-1} + q_0(1 - p_{t-1}) = (q_1 - q_0)p_{t-1} + q_0. \end{aligned}$$

Steady-state: The probabilities no longer change when $p_t = p_{t-1}$,

$$p_t = \frac{q_0}{1 - q_1 + q_0}.$$

Bayes' formula

Bayes' formula relates the two conditional probabilities for a pair of random variables:

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The second equation gives $P(x, y) = P(y \mid x)\, P(x)$. Substituting this into the first,

$$P(x \mid y) = \frac{P(x, y)}{P(y)} = \frac{P(y \mid x)\, P(x)}{P(y)}. \qquad \blacksquare$$

Question

A disease affects 1% of a population. A test is positive 95% of the time if you have the disease, and 10% of the time if you don't. Find $P(D \mid +)$.

Question

It rains on 20% of days. When it rains, the forecast correctly predicts rain 90% of the time. When it doesn't rain, the forecast incorrectly predicts rain 15% of the time. Find $P(\text{rain} \mid \text{forecast rain})$.

Answer key

- $\{1/2, 1/8, 1/8, 1/4\}$: independent? **No**
- 2×2 table $\{1/8, 3/8, 1/8, 3/8\}$: independent? **Yes**
- 2×4 table (1/16s and 3/16s): independent? **Yes**
- 2×4 table (with zeros): independent? **No**
- $P(Y{=}1 \mid X{=}0)$ for $\{3/8, 1/8, 1/4, 1/4\}$: $\mathbf{1/4}$
- $P(X{=}1 \mid Y{=}3)$ for the 2×4 conditional example: $\mathbf{1/3}$
- Conditional model for $\{1/2, 1/8, 1/8, 1/4\}$: $Y \mid X{=}0 \sim \text{Bernoulli}(1/5)$, $Y \mid X{=}1 \sim \text{Bernoulli}(2/3)$
- $X \sim \text{Bernoulli}(1/3)$, $Y \mid X \sim \text{Bernoulli}(X/2 + 1/3)$: joint $= \left(\frac{4}{9}, \frac{2}{9}, \frac{1}{18}, \frac{5}{18}\right)$ for $(0,0), (0,1), (1,0), (1,1)$; marginal $P(Y{=}1) = \mathbf{1/2}$
- Disease test: $P(D \mid +) \approx \mathbf{0.0876}$
- Weather forecast: $P(\text{rain} \mid \text{forecast rain}) = \mathbf{0.6}$